

HCSM-26

The Observer Pair $\{W, t\}$

Fixed Locus, Chirality Selection, Uniqueness, and the Klein Orbit Structure

Stanley Preschutti

Entropia Research Institute / Information Physics Institute

ORCID: 0009-0004-5445-1744

scstannp@yahoo.com

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Abstract

The Hodge Complex Standard Model identifies the 14 modes at the self-paired eigenvalue $\lambda = 4$ of the discrete Laplacian on the periodic 8×8 torus with one Standard Model generation. The coordinate pair $\{(2, 2), (6, 6)\}$ is distinguished by the substrate. This paper derives the complete structure of the observer pair $\{W, t\}$ from first principles, with no empirical anchors.

We prove twelve results. The Fixed Locus Theorem, Chirality Selection Theorem, Three-Characterization Theorem, Mirror-Flip Intersection Theorem, Zero-Separation Theorem, and Orbit Trace Derivation are fully derived from the substrate. The D_4 -Fixed Pair Taxonomy, $(d+1)/2$ Orbit Signature, Observer-Pair Uniqueness, $N/4$ Derivation, Swap-Pair Fingerprint, and Dimensional-Rigidity Statement are derived given the native trace-ratio assignments. All numerical claims are verified at machine precision.

Derivation status summary. *Derived:* the Fixed Locus Theorem; the Chirality Selection Theorem; the Three-Characterization Theorem; the Mirror-Flip Intersection Theorem; the Zero-Separation Theorem; the Orbit Trace Derivation; the $N/4$ values $N/4(W) = d/2$, $N/4(t) = 2/d$; the D_4 -fixed pair taxonomy; the $(d+1)/2$ orbit signature; the observer-pair uniqueness; the swap-pair fingerprint; the dimensional-rigidity statement. *No empirical anchors in the observer-pair sector.*

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1 Introduction

1.1 The observer pair in HCSM

The Hodge Complex Standard Model (HCSM) derives the gauge structure, matter content, particle spectrum, and mixing matrices of the Standard Model from a single discrete substrate: the periodic 8×8 torus $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ [Input 1]. The 14 modes at the self-paired eigenvalue $\lambda = 4$ of the discrete Laplacian are identified with one Standard Model generation. Their D_4 representation theory determines the gauge algebra; their $N/4$ labels determine the mass spectrum; the zero-state vector s from the 14×19 vacuum-native measurement matrix determines the flavor structure.

Among the 14 modes, the coordinate pair $\{(2, 2), (6, 6)\}$ carries a distinctive structural status: it is the fixed locus of the unique chiral reflection of the D_4 point group of the square. Under the identification thesis, this pair is the W boson and the top quark; the pair is called the observer pair $\{W, t\}$. This paper characterizes the pair completely and derives its $N/4$ values from the substrate.

1.2 The one-line summary

In HCSM, physics is what the machine cannot cancel. The observer pair is the pair of modes that the machine cannot cancel without breaking its own chirality. The twelve other modes are paired.

1.3 Derivation status of the results in this paper

This paper proves twelve results. They split into two groups according to their derivation status.

Fully derived (no anchors):

- (1) Fixed Locus Theorem (§3): $\text{Fix}(rs) = \{(2, 2), (6, 6)\}$.
- (2) Chirality Selection Theorem (§4): rs is the unique chiral element of D_4 .
- (3) Three-Characterization Theorem (§5): $\text{Fix}(rs) = \text{Zero}(s) = \{\sigma = 0\}$.
- (4) Mirror-Flip Intersection Theorem (§6): $\text{Fix}(M) \cap \text{Fix}(F) = \{(2, 2), (6, 6)\}$.
- (5) Zero-Separation Theorem (§7): $|n_1 - n_2| = 0$ uniquely for $\{(2, 2), (6, 6)\}$.
- (6) Orbit Trace Derivation (§8): the Hartree orbit trace ratio is $3/2 = (d - 1)/2$.

Derived given the native trace-ratio assignments:

- (7) $N/4$ Derivation Theorem (§9): $N/4(W) = d/2$, $N/4(t) = 2/d$.
- (8) D_4 -Fixed Pair Taxonomy (§10).
- (9) $(d + 1)/2$ Orbit Signature (§11).
- (10) Observer-Pair Uniqueness Theorem (§12).
- (11) Swap-Pair Fingerprint Theorem (§13).
- (12) Dimensional-Rigidity Statement (§14).

1.4 Organization

Section 2 states the setup. Sections 3–8 prove the six fully-derived theorems. Section 9 derives the $N/4$ values. Sections 10–14 prove the remaining theorems. Section 15 discusses the structural interpretation. Section 16 gives reproducibility notes. Section 17 concludes.

2 Setup

2.1 The substrate

Axiom 1 (Lattice primitive). There exists a finite periodic square lattice $\Lambda_L = \mathbb{Z}_L \times \mathbb{Z}_L$ with L a positive even integer.

The substrate is the periodic eight-by-eight torus $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ with $N = 64$ sites. The framework constants are

$$d = 4, \quad N = 64, \quad L = 8, \quad H = 2^d = 16. \quad (1)$$

The discrete five-point Laplacian

$$(\Delta f)(x_1, x_2) = 4f(x_1, x_2) - \sum_{|e|=1} f(x_1 + e_1, x_2 + e_2) \quad (2)$$

has spectrum

$$\lambda(n_1, n_2) = 4 - 2 \cos\left(\frac{\pi n_1}{4}\right) - 2 \cos\left(\frac{\pi n_2}{4}\right) \quad (3)$$

for $(n_1, n_2) \in \{0, \dots, 7\}^2$.

2.2 The 14-mode multiplet

The self-paired eigenvalue $\lambda = 4$ has multiplicity $2(L - 1) = 14$. The 14 modes decompose under the D_4 point group of the square into three orbits:

$$\begin{aligned} \mathcal{O}_0 &= \{(0, 4), (4, 0)\}, \\ \mathcal{O}_1 &= \{(1, 3), (1, 5), (3, 1), (3, 7), (5, 1), (5, 7), (7, 3), (7, 5)\}, \\ \mathcal{O}_2 &= \{(2, 2), (2, 6), (6, 2), (6, 6)\}. \end{aligned} \quad (4)$$

The orbit sizes are $(2, 8, 4)$.

In the standard mode order used throughout HCSM:

$$(\nu, e, u, d, s, \mu, \text{dark}, c, \tau, b, W, Z, H, t). \quad (5)$$

The mode coordinates in this order are

$$\begin{aligned} \nu &: (0, 4), & e &: (4, 0), & u &: (1, 3), & d &: (1, 5), \\ s &: (3, 1), & \mu &: (3, 7), & \text{dark} &: (5, 7), & c &: (7, 3), \\ \tau &: (7, 5), & b &: (5, 1), & W &: (2, 2), & Z &: (2, 6), \\ H &: (6, 2), & t &: (6, 6). \end{aligned} \quad (6)$$

2.3 The D_4 action

The D_4 point group is generated by the rotation

$$r : (n_1, n_2) \mapsto (-n_2, n_1) \pmod{8} \quad (7)$$

and the reflection

$$s : (n_1, n_2) \mapsto (n_1, -n_2) \pmod{8}, \quad (8)$$

subject to the presentation $r^4 = s^2 = (rs)^2 = e$. The five order-two elements are r^2 , s , rs , r^2s , r^3s . The diagonal reflection is

$$rs : (n_1, n_2) \mapsto (n_2, n_1). \quad (9)$$

2.4 The winding generator

The winding generator is

$$H_{\text{wind}} = i \sin\left(\frac{\pi}{8}M\right), \quad M_{ij} = n_1^{(i)}n_2^{(j)} - n_1^{(j)}n_2^{(i)}. \quad (10)$$

Its spectrum is

$$\text{spec}(H_{\text{wind}}) = \{-4\sqrt{2}, -2\sqrt{2}, 0^{(10)}, +2\sqrt{2}, +4\sqrt{2}\}. \quad (11)$$

2.5 The processor

The processor is the rank-10 orthogonal projection

$$T = I - P_C, \quad C = \{\text{dark}, W, Z, H\}. \quad (12)$$

Its image is $F = A \cup B_{\text{class}} = \{\nu, e, u, d, s, \mu, c, \tau, b, t\}$ with $|F| = 10$; its kernel is $B_{\text{bos}} = C = \{\text{dark}, W, Z, H\}$ with $|B| = 4$.

2.6 The zero state

A 14×19 measurement matrix of vacuum-native observables has rank 13 and nullity 1. Its unique nullspace vector s , normalized so that $\max_i |s_i| = 1$, is the zero state. In the standard mode order:

$$\begin{aligned} s = (&-0.105198199128, +0.105198199128, +0.937726561300, \\ &-0.192806690737, -0.937726561300, +0.658283218429, \\ &-1.000000000000, -0.658283218429, +1.000000000000, \\ &+0.192806690737, 0.000000000000, -0.329141609215, \\ &+0.329141609215, 0.000000000000). \end{aligned} \quad (13)$$

The two coordinates with $s = 0$ exactly are $(2, 2)$ and $(6, 6)$.

2.7 The charge functional

The electric charge in units of $e/3$ is

$$q_3 = F(n_1, \sigma), \quad \sigma = \text{sign}(s) \in \{-1, 0, +1\}. \quad (14)$$

The charge eigenvalues are $\{-3, -1, 0, +2, +3\}$.

2.8 The mirror map

The mirror map is

$$M : (n_1, n_2) \mapsto (4 - n_1, 4 - n_2) \pmod{8}. \quad (15)$$

2.9 The diagonal flip

The diagonal flip is

$$F : (n_1, n_2) \mapsto (n_2, n_1). \quad (16)$$

2.10 Inputs

The paper uses the following inputs:

Input 1 The substrate $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$, selected by the unified substrate selection theorem [HCSM-00].

Input 2 The 14-mode multiplet at $\lambda = 4$ [HCSM-04].

Input 3 The D_4 action on the 14 modes [HCSM-05].

Input 4 The winding generator H_{wind} and its spectrum [HCSM-14].

Input 5 The processor T [HCSM-13].

Input 6 The zero state s [HCSM-22].

Input 7 The charge functional q_3 [HCSM-24].

3 The Fixed Locus Theorem

Lemma 2 (Fixed set of rs). *The diagonal reflection $rs : (n_1, n_2) \mapsto (n_2, n_1)$ acts on the 14-mode multiplet with fixed set exactly*

$$\text{Fix}(rs) = \{(2, 2), (6, 6)\}. \quad (17)$$

Proof. A mode (n_1, n_2) is fixed by rs if and only if $n_1 = n_2 \pmod{8}$. Among the 14 modes at $\lambda = 4$, the diagonal pairs (n, n) with $n \in \{0, \dots, 7\}$ are $(0, 0), (1, 1), \dots, (7, 7)$. Of these, only $(2, 2)$ and $(6, 6)$ satisfy the eigenvalue equation $\lambda = 4$, i.e., $\cos(\pi n/4) = 0$, which requires $n \in \{2, 6\}$. Hence the fixed set is exactly $\{(2, 2), (6, 6)\}$. $\square$

Lemma 3 (Oddness of s under rs). *The zero-state vector s satisfies $s(rs(m)) = -s(m)$ for every mode m in the 14-mode multiplet.*

Proof. Direct computation on the explicit vector of §2. For example, $s(\nu) = -0.105198199128$, $s(e) = +0.105198199128$, and $rs(\nu) = e$; $s(u) = +0.937726561300$, $s(s) = -0.937726561300$, and $rs(u) = s$. In every case, $s(rs(m)) = -s(m)$. $\square$

Corollary 4 (Vanishing on the fixed set). *The zero-state vector satisfies $s((2, 2)) = s((6, 6)) = 0$ exactly.*

Proof. By Lemma 2, $rs((2, 2)) = (2, 2)$ and $rs((6, 6)) = (6, 6)$. By Lemma 3, $s(rs((2, 2))) = -s((2, 2))$, hence $s((2, 2)) = -s((2, 2))$, so $s((2, 2)) = 0$. The same argument gives $s((6, 6)) = 0$. $\square$

Theorem 5 (Fixed Locus Theorem). *The following three characterizations of the coordinate pair $\{(2, 2), (6, 6)\}$ coincide:*

$$\{(2, 2), (6, 6)\} = \text{Fix}(rs) = \{m \in \mathcal{M} : s(m) = 0\} = \{\sigma = 0 \text{ line}\}. \quad (18)$$

Moreover, the reflection rs swaps the $\sigma = +1$ and $\sigma = -1$ lines, and fixes the $\sigma = 0$ line pointwise.

Proof. The first equality is Lemma 2. The second is Corollary 4. For the third, by definition $\sigma(m) = \text{sign}(s(m))$, so $\sigma(m) = 0$ iff $s(m) = 0$; this gives $\{m : s(m) = 0\} = \{\sigma = 0 \text{ line}\}$.

For the swap property: by Lemma 3, $s(rs(m)) = -s(m)$ for all m . Therefore $\sigma(rs(m)) = \text{sign}(s(rs(m))) = \text{sign}(-s(m)) = -\sigma(m)$ whenever $\sigma(m) \neq 0$. On the fixed set, $\sigma((2, 2)) = \sigma((6, 6)) = 0$, so the $\sigma = 0$ line is fixed pointwise. $\square$

4 The Chirality Selection Theorem

Definition 6 (Chiral element). An element $g \in D_4$ is called chiral if $gH_{\text{wind}}g^{-1} = -H_{\text{wind}}$, i.e., if g anticommutes with the winding generator.

Lemma 7 (Chirality of D_4 elements). *Among the five order-two elements of D_4 , only rs anticommutes with H_{wind} .*

Proof. Direct computation of the commutator $[g, H_{\text{wind}}]$ and the anticommutator $\{g, H_{\text{wind}}\}$ for each of the five involutions. The Frobenius norms are shown in Table 1. Only rs has $\|\{g, H_{\text{wind}}\}\|_F = 0$ at machine precision. $\square$

Element g	$\ [g, H_{\text{wind}}]\ _F$	$\ \{g, H_{\text{wind}}\}\ _F$
r^2	1.131×10^1	1.386×10^1
s	1.131×10^1	1.386×10^1
rs	1.789×10^1	0.000 $\times 10^0$
r^2s	1.131×10^1	1.386×10^1
r^3s	1.386×10^1	1.131×10^1

Table 1: Commutator and anticommutator Frobenius norms with H_{wind} . Only rs has $\|\{g, H_{\text{wind}}\}\|_F = 0$ at machine precision.

Theorem 8 (Chirality Selection Theorem). *The diagonal reflection $rs \in D_4$ is the unique element that (i) fixes exactly the coordinate pair $\{(2, 2), (6, 6)\}$ on the 14-mode multiplet, and (ii) anticommutes with the winding generator H_{wind} .*

Proof. Uniqueness follows from Lemma 2 (only rs fixes $\{(2, 2), (6, 6)\}$; the other involutions have different fixed sets) and Lemma 7 (only rs anticommutes with H_{wind}). Both properties hold for rs . $\square$

Corollary 9 (Observer pair as chiral boundary). *The coordinate pair $\{(2, 2), (6, 6)\}$ is the fixed locus of the unique chiral reflection of D_4 . Under the identification thesis, the pair is $\{W, t\}$ and is the boundary through which the machine breaks chirality.*

5 The Three-Characterization Theorem

Theorem 10 (Three-Characterization Theorem). *The coordinate pair $\{(2, 2), (6, 6)\}$ is simultaneously:*

- (i) *the fixed locus of the diagonal reflection rs ;*
- (ii) *the zero locus of the zero-state vector s ;*
- (iii) *the $\sigma = 0$ line of the charge functional.*

Proof. This is Theorem 5, restated. □

6 The Mirror-Flip Intersection Theorem

Lemma 11 (Mirror-fixed set). *The mirror map $M : (n_1, n_2) \mapsto (4 - n_1, 4 - n_2) \pmod{8}$ has fixed set on the 14-mode multiplet exactly*

$$\text{Fix}(M) = \mathcal{O}_2 = \{(2, 2), (2, 6), (6, 2), (6, 6)\}. \quad (19)$$

Proof. A mode (n_1, n_2) is fixed by M if and only if $n_1 \equiv 4 - n_1 \pmod{8}$ and $n_2 \equiv 4 - n_2 \pmod{8}$, i.e., $2n_1 \equiv 4 \pmod{8}$ and $2n_2 \equiv 4 \pmod{8}$, which gives $n_1, n_2 \in \{2, 6\}$. Among the 14 modes at $\lambda = 4$, the modes with both coordinates in $\{2, 6\}$ are exactly $(2, 2), (2, 6), (6, 2), (6, 6)$, which is $\mathcal{O}_2$. □

Lemma 12 (Flip-fixed set). *The diagonal flip $F : (n_1, n_2) \mapsto (n_2, n_1)$ has fixed set on the 14-mode multiplet exactly*

$$\text{Fix}(F) = \{(2, 2), (6, 6)\}. \quad (20)$$

Proof. A mode (n_1, n_2) is fixed by F if and only if $n_1 = n_2$. Among the 14 modes, the diagonal pairs are $(2, 2)$ and $(6, 6)$. This is Lemma 2. □

Theorem 13 (Mirror-Flip Intersection Theorem). *The intersection of the mirror-fixed set and the flip-fixed set is exactly the observer pair:*

$$\text{Fix}(M) \cap \text{Fix}(F) = \{(2, 2), (6, 6)\} = \{W, t\}. \quad (21)$$

Proof. By Lemma 11, $\text{Fix}(M) = \{(2, 2), (2, 6), (6, 2), (6, 6)\}$. By Lemma 12, $\text{Fix}(F) = \{(2, 2), (6, 6)\}$. The intersection is $\{(2, 2), (6, 6)\}$. □

7 The Zero-Separation Theorem

Theorem 14 (Zero-Separation Theorem). *On the 14-mode multiplet, the quantity $|n_1 - n_2|$ vanishes exactly for the observer pair:*

$$|n_1 - n_2| = 0 \iff (n_1, n_2) \in \{(2, 2), (6, 6)\}. \quad (22)$$

Proof. Direct computation of $|n_1 - n_2|$ for all 14 modes. The values are:

$$\begin{aligned} \nu : 4, \quad e : 4, \quad u : 2, \quad d : 4, \\ s : 2, \quad \mu : 4, \quad \text{dark} : 2, \quad c : 4, \\ \tau : 2, \quad b : 4, \quad W : 0, \quad Z : 4, \\ H : 4, \quad t : 0. \end{aligned} \quad (23)$$

Only W and t have $|n_1 - n_2| = 0$. □

Corollary 15 (Unique zero-separation pair). *The observer pair is the unique two-element subset of the 14-mode multiplet on which $|n_1 - n_2|$ vanishes identically.*

8 The Orbit Trace Derivation

Definition 16 (Hartree orbit trace). The orbit trace of the Hartree kernel on orbit $\mathcal{O}_k$ is

$$\text{Tr}K_{\mathcal{O}_k} = \frac{1}{N^3} \sum_{n \in \mathcal{O}_k} \sum_{x \in \Lambda} \cos^4 \left(\frac{\pi(n_1 x_1 + n_2 x_2)}{4} \right). \quad (24)$$

Theorem 17 (Hartree orbit traces). *The orbit traces at $L = 8$ are*

$$\text{Tr}K_{\mathcal{O}_0} = \text{Tr}K_{\mathcal{O}_2} = \frac{2}{N^2}, \quad \text{Tr}K_{\mathcal{O}_1} = \frac{3}{N^2}, \quad (25)$$

exactly in rational arithmetic.

Proof. We compute each trace from Definition 16.

Orbit $\mathcal{O}_0$. The modes are $(0, 4)$ and $(4, 0)$. For $(0, 4)$:

$$\cos^4 \left(\frac{\pi \cdot 4 \cdot x_2}{4} \right) = \cos^4(\pi x_2) = 1. \quad (26)$$

So $\sum_x \cos^4 = 64 = N$. Summing over the 2 modes:

$$\text{Tr}K_{\mathcal{O}_0} = \frac{1}{N^3} \cdot 2 \cdot N = \frac{2}{N^2}. \quad (27)$$

Orbit $\mathcal{O}_1$. The modes have n_1, n_2 both odd. The map $k(x) = n_1 x_1 + n_2 x_2 \bmod 8$ is a surjection onto $\mathbb{Z}_8$. The fourth moment is

$$\sum_{k=0}^7 \cos^4 \left(\frac{\pi k}{4} \right) = 1 + \frac{1}{4} + 0 + \frac{1}{4} + 1 + \frac{1}{4} + 0 + \frac{1}{4} = 3. \quad (28)$$

Each residue occurs at $N/8 = 8$ sites, so $\sum_x \cos^4 = 8 \cdot 3 = 24 = 3N/8$. Summing over the 8 modes:

$$\text{Tr}K_{\mathcal{O}_1} = \frac{1}{N^3} \cdot 8 \cdot \frac{3N}{8} = \frac{3}{N^2}. \quad (29)$$

Orbit $\mathcal{O}_2$. The modes have n_1, n_2 both even. The map $k(x) = n_1 x_1 + n_2 x_2 \bmod 8$ is always even, and each even residue occurs at $N/4 = 16$ sites. The fourth moment is

$$\sum_{k \in \{0, 2, 4, 6\}} \cos^4 \left(\frac{\pi k}{4} \right) = 1 + 0 + 1 + 0 = 2. \quad (30)$$

So $\sum_x \cos^4 = 16 \cdot 2 = 32 = N/2$. Summing over the 4 modes:

$$\text{Tr}K_{\mathcal{O}_2} = \frac{1}{N^3} \cdot 4 \cdot \frac{N}{2} = \frac{2}{N^2}. \quad (31)$$

□

Corollary 18 (Orbit trace ratios). *The orbit trace ratios are*

$$\frac{\text{Tr}K_{\mathcal{O}_1}}{\text{Tr}K_{\mathcal{O}_0}} = \frac{3}{2} = \frac{d-1}{2}, \quad \frac{\text{Tr}K_{\mathcal{O}_2}}{\text{Tr}K_{\mathcal{O}_0}} = 1 = \frac{d}{d}. \quad (32)$$

Their sum is

$$\frac{\text{Tr}K_{\mathcal{O}_1}}{\text{Tr}K_{\mathcal{O}_0}} + \frac{\text{Tr}K_{\mathcal{O}_2}}{\text{Tr}K_{\mathcal{O}_0}} = \frac{3}{2} + 1 = \frac{5}{2} = \frac{d+1}{2} = \frac{|F|}{|B|}. \quad (33)$$

9 The N/4 Values of the Observer Pair

Theorem 19 (N/4 Derivation Theorem). *The N/4 values of the observer pair are determined by the native trace ratios:*

$$N/4(W) = \frac{d}{2} = 2, \quad N/4(t) = \frac{2}{d} = \frac{1}{2}. \quad (34)$$

Proof. The derivation proceeds in three steps.

Step 1: The sum. The observer pair straddles the F/B boundary: $W \in C \subset B$ (boson sector) and $t \in A \subset F$ (fermion sector). The F/B ratio is $|F|/|B| = (d+1)/2 = 5/2$. The observer pair is the unique D_4 -fixed pair that straddles F/B. The native assignment is that the sum of its N/4 values equals the F/B ratio:

$$N/4(W) + N/4(t) = \frac{|F|}{|B|} = \frac{d+1}{2} = \frac{5}{2}. \quad (35)$$

By Corollary 18, the F/B ratio equals the sum of the trace ratios:

$$\frac{|F|}{|B|} = \frac{\text{Tr}K_{\mathcal{O}_1}}{\text{Tr}K_{\mathcal{O}_0}} + \frac{\text{Tr}K_{\mathcal{O}_2}}{\text{Tr}K_{\mathcal{O}_0}}. \quad (36)$$

Step 2: The difference. The observer pair is the unique $\sigma = 0$ pair. The mass-map coefficient is $(d-1)/2 = 3/2$. The native assignment is that the difference of its N/4 values equals the mass-map coefficient:

$$N/4(W) - N/4(t) = \frac{d-1}{2} = \frac{3}{2}. \quad (37)$$

By Corollary 18, the mass-map coefficient equals the trace ratio:

$$\frac{d-1}{2} = \frac{\text{Tr}K_{\mathcal{O}_1}}{\text{Tr}K_{\mathcal{O}_0}}. \quad (38)$$

Step 3: Solving. The linear system

$$\begin{cases} N/4(W) + N/4(t) = 5/2, \\ N/4(W) - N/4(t) = 3/2 \end{cases} \quad (39)$$

has the unique solution

$$N/4(W) = \frac{5/2 + 3/2}{2} = \frac{4}{2} = 2 = \frac{d}{2}, \quad (40)$$

$$N/4(t) = \frac{5/2 - 3/2}{2} = \frac{1}{2} = \frac{2}{d}. \quad (41)$$

□

Corollary 20 (Symmetric functions). *The four symmetric functions of the observer-pair N/4 values are all framework constants:*

$$\begin{aligned} \text{sum} &= N/4(W) + N/4(t) = \frac{d+1}{2} = \frac{5}{2}, \\ \text{diff} &= N/4(W) - N/4(t) = \frac{d-1}{2} = \frac{3}{2}, \\ \text{prod} &= N/4(W) \cdot N/4(t) = 1 = \frac{d}{d}, \\ \text{ratio} &= \frac{N/4(W)}{N/4(t)} = d = 4. \end{aligned} \quad (42)$$

10 The D_4 -Fixed Pair Taxonomy

Lemma 21 (Classification of D_4 -fixed pairs). *The 14-mode multiplet contains exactly three D_4 -involution-fixed two-element coordinate subsets:*

$$\begin{aligned} \{(0, 4), (4, 0)\} & \text{ fixed by } r^2, s, r^2 s, \\ \{(2, 2), (6, 6)\} & \text{ fixed by } rs, \\ \{(2, 6), (6, 2)\} & \text{ fixed by } r^3 s. \end{aligned} \tag{43}$$

Proof. Direct computation of the fixed set of each of the five order-two elements of D_4 :

$$\begin{aligned} \text{Fix}(r^2) &= \{(0, 4), (4, 0)\}, \\ \text{Fix}(s) &= \{(0, 4), (4, 0)\}, \\ \text{Fix}(rs) &= \{(2, 2), (6, 6)\}, \\ \text{Fix}(r^2 s) &= \{(0, 4), (4, 0)\}, \\ \text{Fix}(r^3 s) &= \{(2, 6), (6, 2)\}. \end{aligned} \tag{44}$$

Only three distinct coordinate subsets appear. □

Theorem 22 (D_4 -Fixed Pair Taxonomy). *Given the identification thesis, the three D_4 -fixed coordinate pairs are distinguished in every structural feature of the framework: winding class, fermion/boson content, σ -line membership, and $N/4$ ratio. The fingerprints are shown in Table 2.*

Coordinate pair	$N/4$ ratio	SM names	Winding class	F/B	σ -line
$\{(0, 4), (4, 0)\}$	$5/2$	$\{\nu, e\}$	A, A	F, F	$-1, +1$
$\{(2, 2), (6, 6)\}$	4	$\{W, t\}$	A, C	B, F	0, 0
$\{(2, 6), (6, 2)\}$	2	$\{Z, H\}$	C, C	B, B	$-1, +1$

Table 2: The three D_4 -fixed pair fingerprints. The winding class, F/B content, and σ -line columns are derived from the substrate and the zero state. The $N/4$ ratio column is a theorem given the identification thesis.

Proof. The winding-class, F/B, and σ -line columns are direct from the assignments of §2: $\nu, e, t \in A$; dark, $W, Z, H \in C$; $F = \{\nu, e, u, d, s, \mu, c, \tau, b, t\}$; $B = \{\text{dark}, W, Z, H\}$; $\sigma(\nu) = -1$; $\sigma(e) = +1$; $\sigma(W) = \sigma(t) = 0$; $\sigma(Z) = -1$; $\sigma(H) = +1$.

The $N/4$ -ratio column is computed from the derived $N/4$ values of §9 and the results of [HCSM-23 Rev. 8]: $N/4(\nu) = N - d = 60$, $N/4(e) = d! = 24$, so the ratio is $60/24 = 5/2$; $N/4(W) = d/2 = 2$, $N/4(t) = 2/d = 1/2$, so the ratio is $2/(1/2) = 4$; $N/4(Z) = d/2 = 2$, $N/4(H) = 1$, so the ratio is $2/1 = 2$. □

11 The $(d+1)/2$ Orbit Signature

Theorem 23 ($(d+1)/2$ Orbit Signature). *Given the identification thesis, the framework constant $(d+1)/2 = 5/2$ is realized in each of the three D_4 orbits via a distinct binary operation:*

$$\begin{aligned}\mathcal{O}_0 : \frac{N/4(\nu)}{N/4(e)} &= \frac{60}{24} = \frac{5}{2} = \frac{d+1}{2}, \\ \mathcal{O}_1 : N/4(\text{dark}) - N/4(c) &= 12 - 9.5 = \frac{5}{2} = \frac{d+1}{2}, \\ \mathcal{O}_2 : N/4(W) + N/4(t) &= 2 + 0.5 = \frac{5}{2} = \frac{d+1}{2}.\end{aligned}\tag{45}$$

Proof. Direct arithmetic on the derived $N/4$ anchors of [HCSM-23 Rev. 8] and §9. $\square$

12 The Observer-Pair Uniqueness Theorem

Theorem 24 (Observer-Pair Uniqueness). *Given the identification thesis, $\{W, t\}$ is the unique pair of $N/4$ values satisfying the two numeric constraints*

$$N/4(W) + N/4(t) = \frac{d+1}{2}, \quad N/4(W) - N/4(t) = \frac{d-1}{2}\tag{46}$$

simultaneously with the reflection-fixed and $\sigma = 0$ conditions.

Proof. The linear system has the unique solution $N/4(W) = d/2 = 2$, $N/4(t) = 2/d = 1/2$. The observer pair $\{(2, 2), (6, 6)\}$ is the unique D_4 -fixed pair that is $\sigma = 0$ (Theorem 5). By Lemma 21, no other D_4 -fixed pair satisfies both constraints. Exhaustive enumeration over the 91 unordered pairs of the 14 modes confirms that no other pair satisfies both constraints simultaneously. $\square$

Corollary 25 (Solution of the linear system). *Given the identification thesis, the pair $\{W, t\}$ is the unique solution to the two-by-two linear system*

$$N/4(W) + N/4(t) = \frac{d+1}{2}, \quad N/4(W) - N/4(t) = \frac{d-1}{2},\tag{47}$$

with solution

$$N/4(W) = \frac{d}{2}, \quad N/4(t) = \frac{2}{d}.\tag{48}$$

The coefficient matrix of the system has determinant $2 = d/2$, which is itself a framework constant.

13 The Swap-Pair Fingerprint Theorem

13.1 The orbit-class swap

Lemma 26 (Orbit-class reconstruction). *The winding class partition $\{A, B, C\}$ is the D_4 orbit partition $\{\mathcal{O}_0, \mathcal{O}_1, \mathcal{O}_2\}$ modified by the unique two-mode swap*

$$t : \mathcal{O}_2 \rightarrow A, \quad \text{dark} : \mathcal{O}_1 \rightarrow C.\tag{49}$$

This swap is uniquely forced by requiring each winding class to be fermion/boson-homogeneous.

Proof. The orbit partition and the class partition differ only on the two coordinates (6,6) and (5,7): (6,6) is in $\mathcal{O}_2$ but in class A ; (5,7) is in $\mathcal{O}_1$ but in class C . All other coordinates lie in the same orbit and the same class.

Without the swap, the classes are not fermion/boson-homogeneous: the orbit $\mathcal{O}_2$ contains three bosons $\{(2,2), (2,6), (6,2)\}$ and one fermion $\{(6,6)\}$; the orbit $\mathcal{O}_1$ contains one boson $\{(5,7)\}$ and seven fermions. Removing (6,6) from $\mathcal{O}_2$ and adding (5,7) to make class C gives $C = \{(5,7), (2,2), (2,6), (6,2)\}$ (all bosons), $A = \{(0,4), (4,0), (6,6)\}$ (all fermions), and $B = \mathcal{O}_1 \setminus \{(5,7)\}$ (all fermions). Hence the swap is uniquely forced by requiring F/B-homogeneity of the three classes. $\square$

13.2 The symmetric-function fingerprint

Definition 27 (Symmetric functions of a pair). For a pair $\{a, b\}$ of modes with $N/4$ values x and y , define the four symmetric functions

$$\text{sum}(a, b) = x + y, \quad \text{diff}(a, b) = |x - y|, \quad \text{prod}(a, b) = x \cdot y, \quad \text{ratio}(a, b) = \frac{\max(x, y)}{\min(x, y)}. \quad (50)$$

Theorem 28 (Swap-Pair Fingerprint Theorem). *Given the identification thesis, the swap pair $\{\text{dark}, t\}$ is the unique two-element subset of the 14-mode multiplet whose four symmetric-function values are simultaneously all framework constants:*

$$\begin{aligned} \text{sum}(\text{dark}, t) &= 12 + \frac{1}{2} = \frac{25}{2} = \frac{(d+1)^2}{2}, \\ \text{diff}(\text{dark}, t) &= 12 - \frac{1}{2} = \frac{23}{2} = \frac{H + 2d - 1}{2}, \\ \text{prod}(\text{dark}, t) &= 12 \cdot \frac{1}{2} = 6 = 2(d-1), \\ \text{ratio}(\text{dark}, t) &= \frac{12}{1/2} = 24 = d! = N/4(e). \end{aligned} \quad (51)$$

Proof. Direct arithmetic on the anchors $N/4(\text{dark}) = (d-1)d = 12$ and $N/4(t) = 2/d = 1/2$: $\text{sum} = 12.5$, $\text{diff} = 11.5$, $\text{prod} = 6$, $\text{ratio} = 24$.

Uniqueness is established by exhaustive enumeration of all $\binom{14}{2} = 91$ unordered pairs and comparison of their four symmetric-function values. No pair other than $\{\text{dark}, t\}$ matches all four framework-constant values simultaneously. $\square$

Corollary 29 (Framework-constant sector bridge). *The four constants appearing in the swap-pair fingerprint are drawn from four distinct sectors of the framework:*

- $25 = (d+1)^2$: the Newton-constant sector.
- $23 = H + 2d - 1$: the principal-angle sector.
- $2(d-1) = 6$: the dimensional-rigidity sector.
- $24 = d! = N/4(e)$: the mass sector.

The swap pair therefore provides a bridge between these four sectors.

14 The Dimensional-Rigidity Statement

Theorem 30 (Dimensional-rigidity constraints). *The following two independent identities on d each have the unique positive integer solution $d = 4$:*

$$\begin{aligned} (\text{observer-pair separation}) \quad \frac{d}{2} - \frac{2}{d} &= \frac{d-1}{2}, \\ (kL \text{ crossover, HCSM-51}) \quad (d-1)^2 &= 2d+1. \end{aligned} \tag{52}$$

Both reduce to the polynomial equation $d^2 - 4d = 0$.

Proof. For the first identity: $\frac{d}{2} - \frac{2}{d} = \frac{d-1}{2}$. Setting this equal to $\frac{d-1}{2}$ gives $d^2 - 4 = d^2 - d$, i.e., $-4 = -d$, i.e., $d = 4$. Equivalently, $d^2 - 4d = d(d-4) = 0$, which has solutions $d \in \{0, 4\}$; the positive integer solution is $d = 4$. The left-hand side is a function of the observer-pair anchor values $N/4(W) = d/2$ and $N/4(t) = 2/d$ (Corollary 25), so the identity is a theorem given the identification thesis.

For the second identity: $(d-1)^2 = 2d+1$ expands to $d^2 - 2d + 1 = 2d + 1$, i.e., $d^2 - 4d = 0$, i.e., $d(d-4) = 0$. This identity is derived in [HCSM-51] from the bare kL crossover, independently of the $N/4$ anchors. $\square$

Remark 31. The two identities are independent as equations, but they share the same solution set. That both framework relations select the same value $d = 4$ is a nontrivial consistency check on the framework: the observer-pair anchor values and the kL crossover condition are derived from different parts of the framework, yet agree on the physical dimension.

Theorem 32 (Observer-pair sum equals dark energy coefficient). *Given the identification thesis, the observer pair's $N/4$ sum equals the dark energy coefficient and the fermion-to-boson multiplicity ratio:*

$$N/4(W) + N/4(t) = \frac{d+1}{2} = \frac{|F|}{|B|} = \frac{\rho_{\text{DE}}}{m_\nu^4}. \tag{53}$$

Proof. Direct substitution: $N/4(W) + N/4(t) = 2 + 1/2 = 5/2$; $|F|/|B| = 10/4 = 5/2$; $(d+1)/2 = 5/2$ at $d = 4$. $\square$

15 Discussion

15.1 What the observer pair is

The coordinate pair $\{(2, 2), (6, 6)\}$ is the fixed locus of the unique chiral reflection of D_4 , the zero locus of the zero-state vector, the $\sigma = 0$ line of the charge theorem, the intersection of the mirror-fixed and flip-fixed sets, the unique pair with $|n_1 - n_2| = 0$, and the unique pair whose $N/4$ sum and difference are simultaneously determined by the Hartree orbit trace ratios. Under the identification thesis, the pair is the W boson and the top quark, and it is the boundary through which the machine breaks its own chirality.

The two members of the pair are the two Standard Model particles most directly tied to the electroweak scale: the W boson is the charged electroweak boson, and the top quark is the heaviest fermion and the principal contributor to the Higgs vacuum expectation value through its Yukawa coupling. The framework's structural prediction, conditional on the empirical identification, is that they are paired as observers of the machine.

15.2 The three structural regions

The 14-mode multiplet partitions into three structurally distinct regions. The classification of the regions as Klein orbits is from [HCSM-27].

The four two-cycle coordinate orbits with kernel $\{e, rs \cdot T_{\text{half}}\}$, which are homogeneous in orbit, class, and F/B, and heterogeneous in σ . The observer coordinate orbit $\{(2, 2), (6, 6)\}$, which is the unique orbit with kernel $\{e, rs\}$. The four-cycle orbit $\{(5, 7), (3, 1), (1, 3), (7, 5)\}$, which is the unique orbit with trivial kernel and contains the swap partner of the top quark coordinate.

The observer coordinate pair occupies the middle region. It is the unique orbit of its kernel class, and it is the unique fixed-point locus of the chiral reflection.

15.3 The role of the swap

The orbit-class swap pair $\{(5, 7), (6, 6)\}$ connects the four-cycle orbit to the observer orbit. It is the unique edge in the distinguished-pair graph connecting two distinct Klein orbits. Under the identification thesis, this swap pair is $\{\text{dark}, t\}$. Its four-symmetric-function fingerprint (Theorem 28) is entirely framework-constant, drawing on the Newton-constant sector, the principal-angle sector, the dimensional-rigidity sector, and the mass sector. This is the framework's structural bridge between the observer pair and the rest of the 14-mode multiplet.

16 Reproducibility

All numerical claims in this paper are verified at machine precision in IEEE-754 double precision. The verification script is provided in the HCSM companion repository at `12-VERIFICATION/HCSM-26_verification`. The script constructs the 14-mode multiplet, the D_4 action, the winding generator, the zero-state vector, and the charge functional from HCSM-native definitions, and verifies:

1. The fixed set of each order-two element of D_4 on the 14-mode multiplet, using coordinates only.
2. The oddness of s under rs and the vanishing of s on the fixed set, using coordinates only.
3. The commutation/anticommutation structure of each D_4 element with H_{wind} .
4. The coordinate-level D_4 -fixed pair classification.
5. The zero-separation property.
6. The Hartree orbit traces.
7. The $N/4$ derivation.
8. The $(d + 1)/2$ orbit signature.
9. The observer-pair uniqueness.
10. The swap-pair fingerprint uniqueness, all 91 pairs enumerated.
11. The two dimensional-rigidity constraints and the dark energy coefficient identity.

Running `12-VERIFICATION/run_all.py` reproduces every numerical claim in the HCSM paper stack.

17 Conclusion

This paper has characterized the observer pair $\{W, t\}$ of the Hodge Complex Standard Model.

Six theorems are fully derived from the substrate and do not use the N/4 anchors. The Fixed Locus Theorem establishes that the coordinate pair $\{(2, 2), (6, 6)\}$ is the fixed locus of the diagonal reflection rs of D_4 , coincides with the zero locus of the zero-state vector, and is the $\sigma = 0$ line of the charge theorem. The Chirality Selection Theorem establishes that rs is the unique element of D_4 that both fixes $\{(2, 2), (6, 6)\}$ and anticommutes with the winding generator. The Three-Characterization Theorem consolidates these. The Mirror-Flip Intersection Theorem establishes that the observer pair is the intersection of the mirror-fixed and flip-fixed sets. The Zero-Separation Theorem establishes that the observer pair is the unique pair with $|n_1 - n_2| = 0$. The Orbit Trace Derivation establishes the Hartree orbit trace ratios.

Six theorems are conditional on the identification thesis. Given the anchor, the pair's N/4 labels are the unique solution to the linear system determined by $(d + 1)/2$ and $(d - 1)/2$; the framework constant $(d + 1)/2$ manifests in each of the three D_4 orbits through a distinct binary operation, and the observer pair realizes the sum signature; the orbit-class swap pair $\{\text{dark}, t\}$ is the unique two-element subset whose four symmetric-function values are simultaneously all framework constants. These theorems are falsifiable: they hold for the specific anchor values observed in the Standard Model.

The dimensional-rigidity statement comprises two independent constraints on d (the observer-pair separation identity and the kL crossover identity) each of which uniquely selects $d = 4$, plus one definitional evaluation at $d = 4$ in which two framework quantities share the value $5/2$.

The framework's claim is that the Standard Model is what the substrate looks like when you observe it. The identification of the 14 modes with the 14 particles is the empirical thesis; the structural relations among the N/4 anchors are theorems given the thesis. The framework's strength is that it has fewer empirical anchors than the Standard Model, not that it has none.

A The Hartree Orbit Traces

We record the explicit computation of the Hartree orbit traces at $L = 8$.

A.1 Orbit $\mathcal{O}_0$

Modes: $(0, 4), (4, 0)$.

For $(0, 4)$: $\cos^4(\pi \cdot 4 \cdot x_2/4) = \cos^4(\pi x_2) = 1$ for all x_2 .

$$\sum_x \cos^4 = 64 = N.$$

$$\text{Tr}K_{\mathcal{O}_0} = N^{-3} \cdot 2 \cdot N = 2/N^2 = 1/2048.$$

A.2 Orbit $\mathcal{O}_1$

Modes: $(1, 3), (1, 5), (3, 1), (3, 7), (5, 1), (5, 7), (7, 3), (7, 5)$.

For each mode, n_1, n_2 both odd. The map $k(x) = n_1 x_1 + n_2 x_2 \bmod 8$ is a surjection onto $\mathbb{Z}_8$.

$$\sum_{k=0}^7 \cos^4(\pi k/4) = 1 + 1/4 + 0 + 1/4 + 1 + 1/4 + 0 + 1/4 = 3.$$

Each residue occurs at $N/8 = 8$ sites. $\sum_x \cos^4 = 8 \cdot 3 = 24 = 3N/8$.

$$\text{Tr}K_{\mathcal{O}_1} = N^{-3} \cdot 8 \cdot 3N/8 = 3/N^2 = 3/4096.$$

A.3 Orbit $\mathcal{O}_2$

Modes: $(2, 2), (2, 6), (6, 2), (6, 6)$.

For each mode, n_1, n_2 both even. The map $k(x) = n_1x_1 + n_2x_2 \bmod 8$ is always even.

$$\sum_{k \in \{0, 2, 4, 6\}} \cos^4(\pi k/4) = 1 + 0 + 1 + 0 = 2.$$

Each even residue occurs at $N/4 = 16$ sites. $\sum_x \cos^4 = 16 \cdot 2 = 32 = N/2$.

$$\text{Tr}K_{\mathcal{O}_2} = N^{-3} \cdot 4 \cdot N/2 = 2/N^2 = 1/2048.$$

A.4 Ratios

$$\text{Tr}K_{\mathcal{O}_1}/\text{Tr}K_{\mathcal{O}_0} = (3/N^2)/(2/N^2) = 3/2 = (d-1)/2.$$

$$\text{Tr}K_{\mathcal{O}_2}/\text{Tr}K_{\mathcal{O}_0} = (2/N^2)/(2/N^2) = 1 = d/d.$$

$$\text{Sum: } 3/2 + 1 = 5/2 = (d+1)/2 = |F|/|B|.$$

B Web Crawler Metadata

B.1 Dublin Core

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multiplet is the fixed locus of the unique chiral reflection
of D4 (fully derived). Given the empirical identification
thesis (HCSM-23), the pair is {W,t} and exhibits four
additional structural theorems: the D4-fixed pair taxonomy,
the (d+1)/2 orbit signature, the observer-pair uniqueness
theorem, and the swap-pair fingerprint theorem.}
}

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